ModelIt!

Student Activity Pack

Ocean Currents and Climate: The Conveyor Belt

The Hidden Rivers That Control Our Weather

Grade 07L2 | NGSS MS-ESS2-6, MS-ESS3-5 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Do you think the ocean moves? If so, what makes it move?

2 Why do you think some coastal cities have mild winters while inland cities at the same latitude have harsh ones?

3 Draw what you think happens to warm water from the equator as it moves toward the poles.

4 How do you think the ocean affects weather on land?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Temperature Difference (Equator vs Poles) | Wind Patterns | Ocean Current Speed | Heat Distribution | Regional Climate Stability

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Normal Circulation

Moderate temperature gradient, normal wind patterns

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Enhanced Temperature Gradient

Maximum difference between equator and poles

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Ice Melt Disruption

Reduced temperature gradient from polar ice melt

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Salinity Levels — The salt concentration in ocean water, which affects water density and drives deep-water currents

Coriolis Effect — The deflection of moving water and air caused by Earth's rotation, which curves ocean currents

Sea Surface Temperature — The temperature of the ocean's surface layer, which affects evaporation and storm formation

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do invisible rivers in the ocean control the weather thousands of miles away? Use evidence from your simulation to support your answers.

1. **Why is England warmer than Labrador, Canada, even though they are at the same latitude?**

2. **How does your model show that ocean currents are driven by temperature differences rather than just wind?**

3. **What would happen to global climate if the Great Ocean Conveyor Belt completely stopped?**

4. **How is the ocean current system an example of Earth's systems interacting with each other?**

5. **Why could global warming paradoxically make some parts of Europe colder?**

8 STEM Challenge: Design a Climate Impact Map

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YOUR ENGINEERING MISSION

Create a visual map showing how ocean current disruption would affect climate in different regions around the world, using evidence from your simulation model.

THE SCENARIO

NOAA scientists are studying what would happen if the Atlantic Meridional Overturning Circulation (AMOC) weakened by 50%. They need your team to create a visual impact map showing which regions would be affected and how.

Engineering Design Thinking

1. HOW WILL YOU REPRESENT OCEAN CURRENT FLOW ON YOUR MAP?
2. WHAT COLOR CODING WILL SHOW TEMPERATURE CHANGES IN DIFFERENT REGIONS?
3. HOW WILL YOU VISUALIZE THE BEFORE AND AFTER OF CURRENT DISRUPTION?
4. WHAT MODEL EVIDENCE WILL YOU CITE TO SUPPORT YOUR IMPACT PREDICTIONS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Physical Oceanographers study ocean currents, waves, and water properties to understand how oceans affect climate and weather. They work for NOAA, universities, and environmental research institutes, earning $70,000-$130,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 London, England, and Goose Bay, Canada, are at nearly the same latitude, yet London's average winter temperature is about 10 degrees Celsius warmer. Based on the ocean circulation model, which factor best explains this difference?

- **A.** London receives more direct sunlight than Goose Bay because of Earth's axial tilt
- **B.** The Gulf Stream carries warm tropical water across the Atlantic to northwestern Europe, delivering heat that moderates London's climate
- **C.** England is a smaller landmass, so it heats up faster than the Canadian mainland
- **D.** London is at a lower elevation than Goose Bay, and lower elevations are always warmer

2 In the model, reducing the Temperature Difference between the equator and poles (simulating polar ice melt) causes Ocean Current Speed to decrease. Which cascade of effects does this trigger?

- **A.** Slower currents increase Heat Distribution, making all regions warmer
- **B.** Slower currents reduce Heat Distribution, causing tropical regions to overheat while polar-adjacent regions lose warmth, destabilizing regional climates
- **C.** Slower currents have no effect on Heat Distribution because wind patterns compensate completely
- **D.** Slower currents cause the Temperature Difference to increase, creating a self-correcting cycle

3 A student claims that global warming should make ALL regions warmer, so Europe cannot get colder. Using the ocean circulation model, which argument best refutes this claim?

- **A.** Global warming only affects tropical regions, not temperate ones
- **B.** Global warming melts polar ice, reducing salinity and the temperature gradient that drives ocean currents. Weakened currents reduce heat delivery to Europe, potentially making it colder even as global averages rise
- **C.** Europe is protected from global warming by the Atlantic Ocean
- **D.** Climate models cannot predict regional effects because weather is too chaotic

4 The model shows that Heat Distribution has a negative relationship with Temperature Difference. This means that as ocean currents distribute heat more evenly, the temperature gradient that drives them weakens. What type of mechanism does this represent?

- **A.** A positive feedback loop that amplifies ocean current speed indefinitely
- **B.** A self-regulating negative feedback loop where the system moderates its own driving force
- **C.** A one-directional cause-and-effect chain with no feedback
- **D.** An error in the model because outputs cannot affect inputs

5 **In the model, both Temperature Difference and Wind Patterns have positive relationships with Ocean Current Speed. A student wants to determine which factor has a greater influence on current speed. Which investigation approach would best answer this question?**

- **A.** Set both inputs to maximum and observe the result
- **B.** Change one variable at a time while holding the other constant, then compare the resulting changes in Ocean Current Speed
- **C.** Remove both inputs and observe whether currents still flow
- **D.** Only test extreme values because moderate values do not reveal meaningful patterns